The Social Media Problem Space and Paths to Improvement

Bao T. Truong · 29.10.2026

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SynoSys

House keeping

- Website? https://btrantruong.github.io/algo-exp-research-course/
- Deadline for forming group
- Reading academic paper

Objectives

- Understand how platform design and user behavior co-produce social-media outcomes, and
- how causal research can diagnose and improve them.

Tell us about your social media experience

- What do you like about it?
- What do you dislike?
- What would make the experience better?

Social media Ig, Tiktok, X

😀

Connecting w friends
Inspirations
Interest
News
Free (monetary)

😕

Consumerism
Privacy (tracking for personalization, access to personal info)

Addictive
Stuck in your bubble
Bots & trolls
Bullying & mobbing
Misleading
Fake engagement
Lack of diversity
Ragebait
Slop

🤔🤔

Strict privacy laws

More human monitoring

More diverse recommend (tradeoff for relevance)

Verification & Bot detection

Why do these feelings persist?

Traditional media: one-to-many



A small group of editors and publishers selects content for a large, shared audience.

Editors are gatekeepers



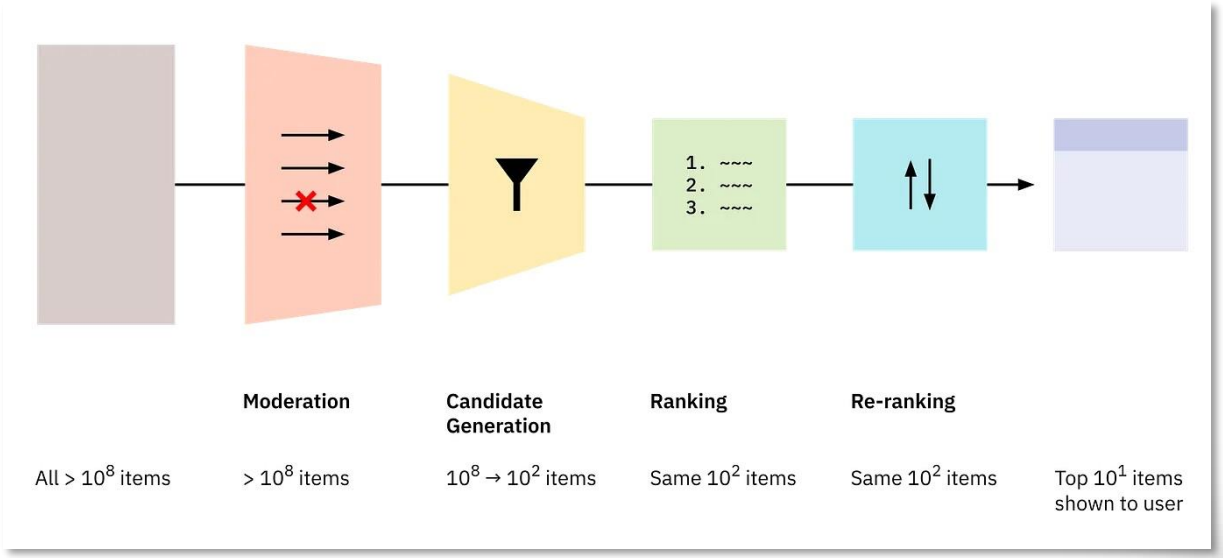
Social media: many-to-many



The scale of information is simply impossible for humans



- NYT produces ~144 articles/day; a small editorial team can curate a shared homepage.
- Social media creates millions of potential posts every day.
- If 5 editors spent 2 minutes per post, reviewing 10M daily posts would take ~23 years.





How to go from 10 million posts to a few posts in 200ms?



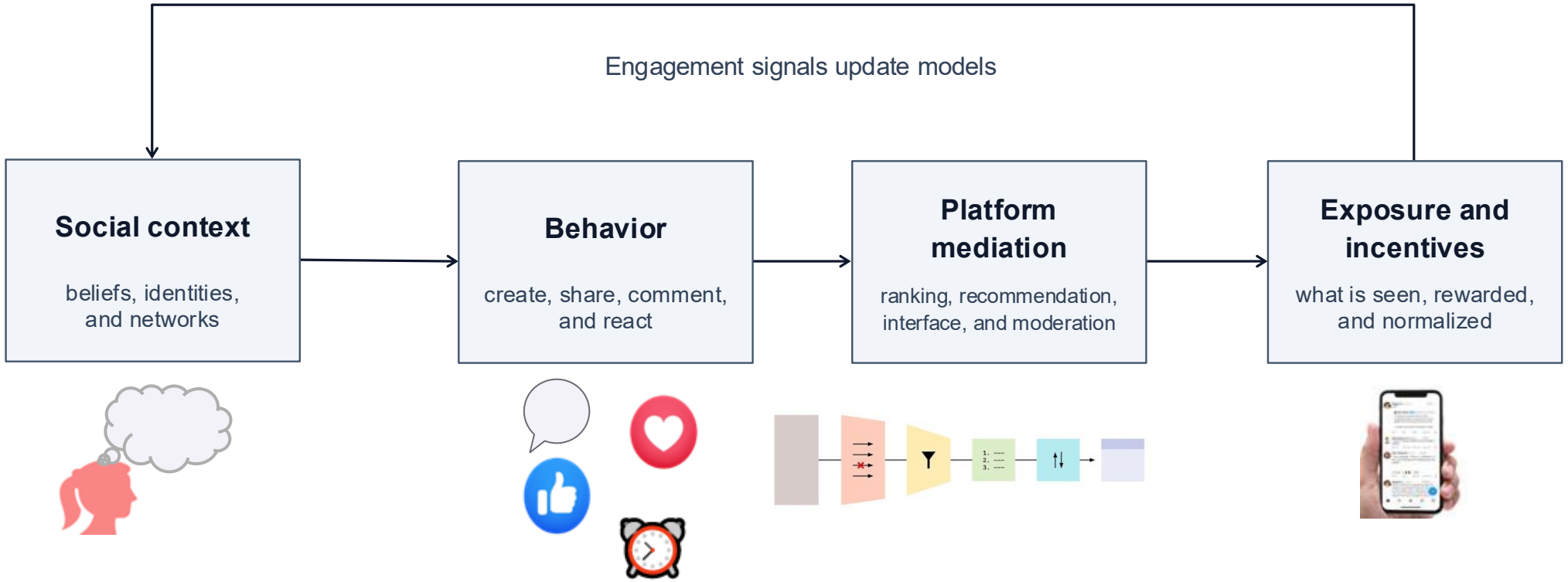
Feedback loop

Social context

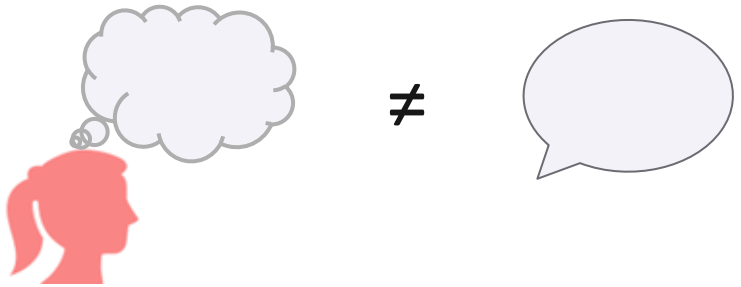
beliefs, identities,
and networks



Feedback loop



Individual cognitive biases



Confirmation bias - People have a tendency to seek out, notice, interpret, and remember information in ways that support what we already believe.

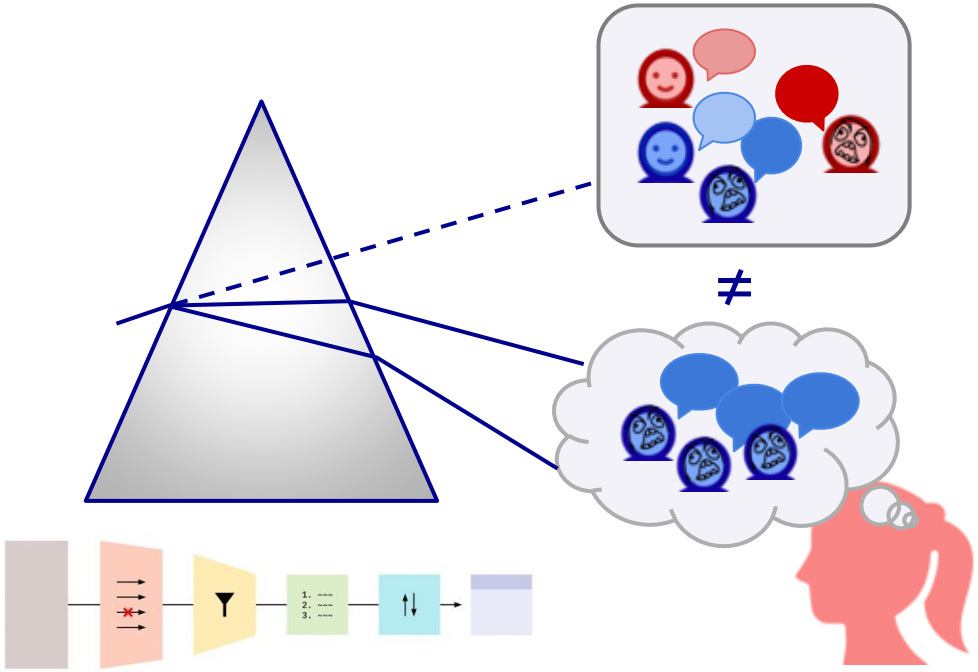
Bandwagon effect - People are more likely to adopt a belief, buy something, or support a candidate because they see that many other people already do. Popularity itself becomes persuasive: “If everyone supports it, it must be right.”

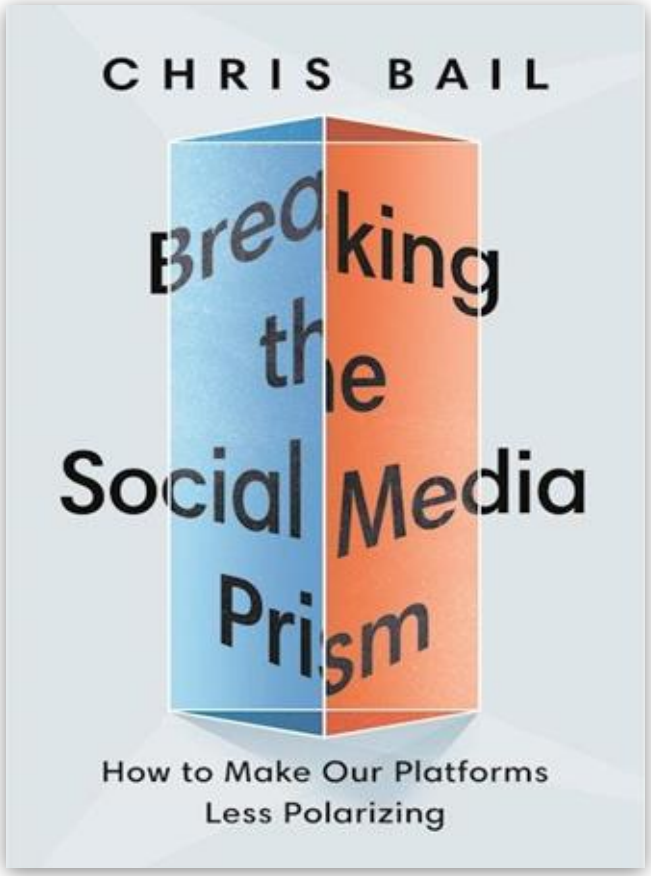
Mere exposure effect - People tend to develop greater familiarity and often greater liking for something simply because they encounter it repeatedly. Repeated exposure to a slogan, face, brand, or political claim can make it feel more familiar and therefore more acceptable.



Social biases

Homophily The tendency for people to connect with, interact with, or form relationships with others who are similar to themselves.





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Political segregation observed in retweet network

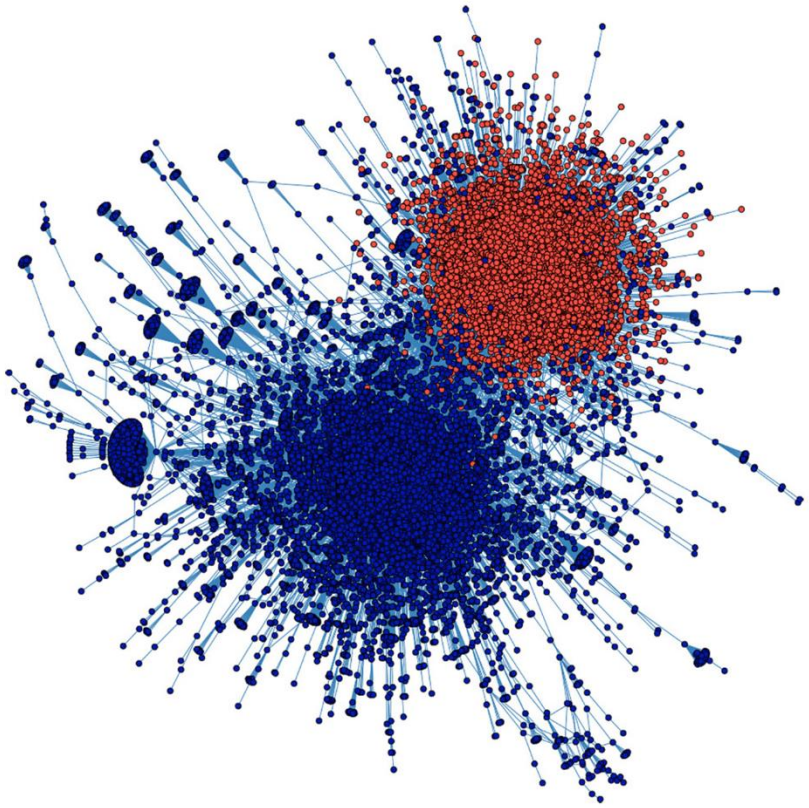


Figure 3 The network of political retweets, laid out using a force-directed algorithm. Node colors

2010

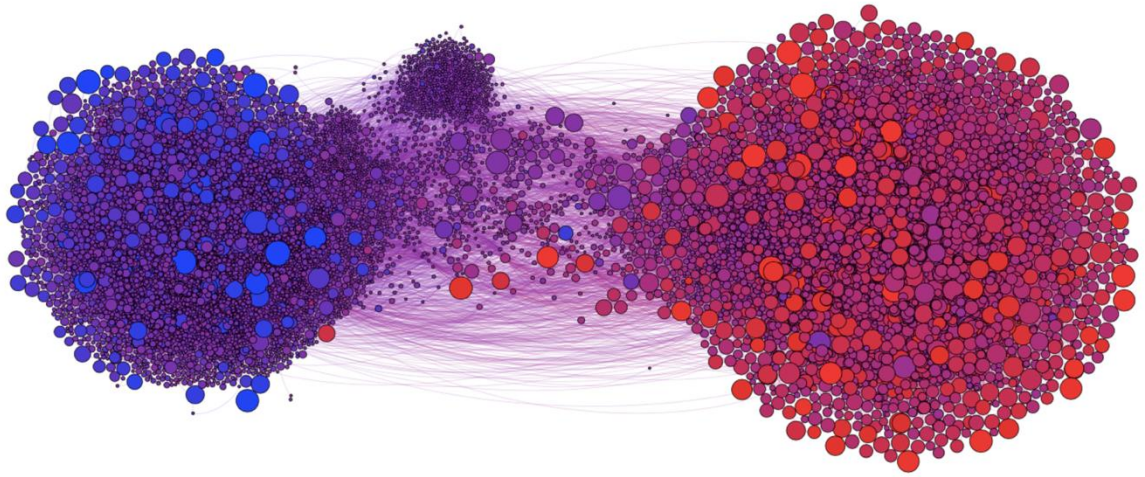


IMAGE: ECHO-CHAMBER STRUCTURE OF DIFFUSION NETWORK. NODE COLOR AND SIZE ARE BASED ON PARTISANSHIP AND MISINFORMATION, RESPECTIVELY

2016

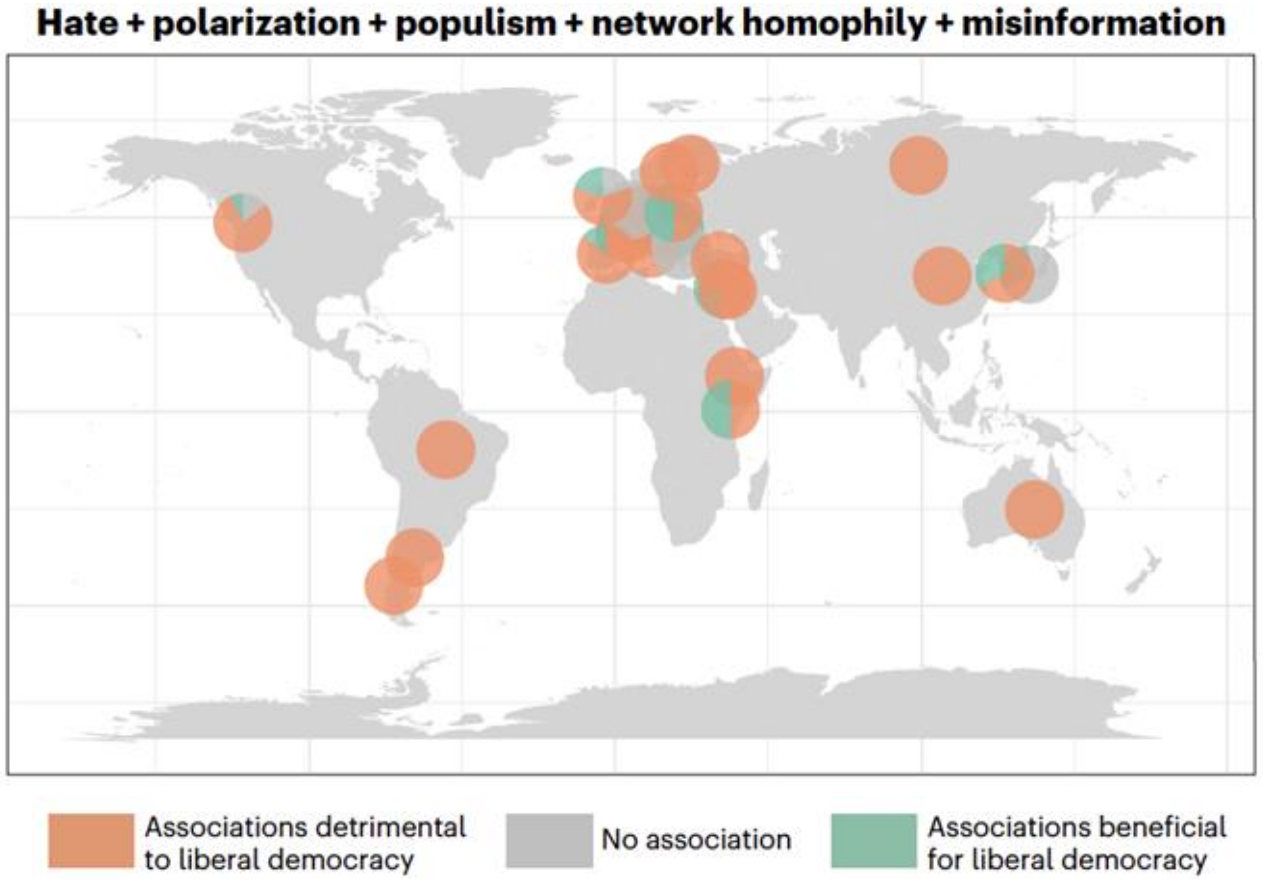
Conover, M., et al. Political polarization on twitter. 2011.
Nikolov, D., et al. Right and left, partisanship predicts (asymmetric) vulnerability to misinformation. 2016

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Negative outcomes

- Hate speech
- Misinformation
- Polarization
- Steoreotypes



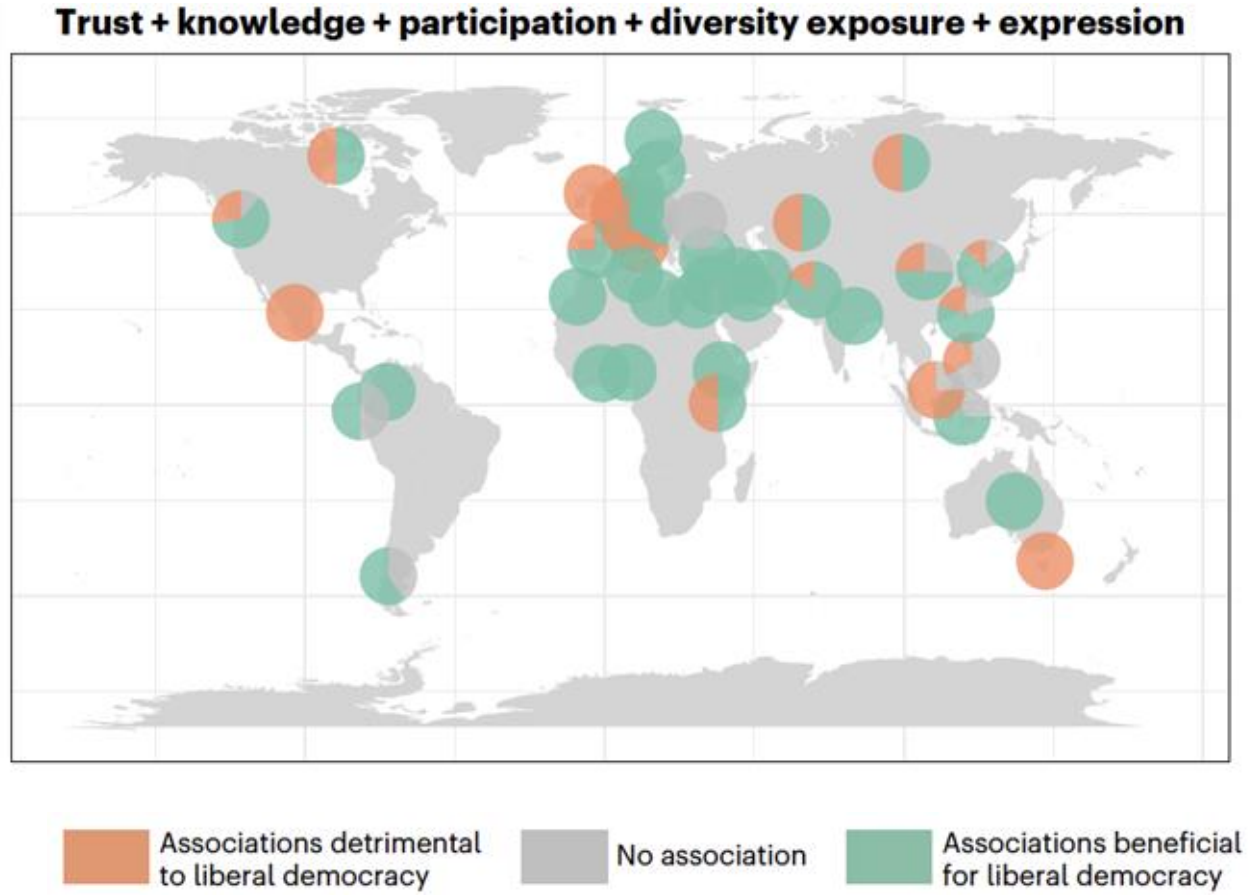
Lorenz-Spreen, P. *et al.* A systematic review of worldwide causal and correlational evidence on digital media and democracy. 2023

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Positive outcomes

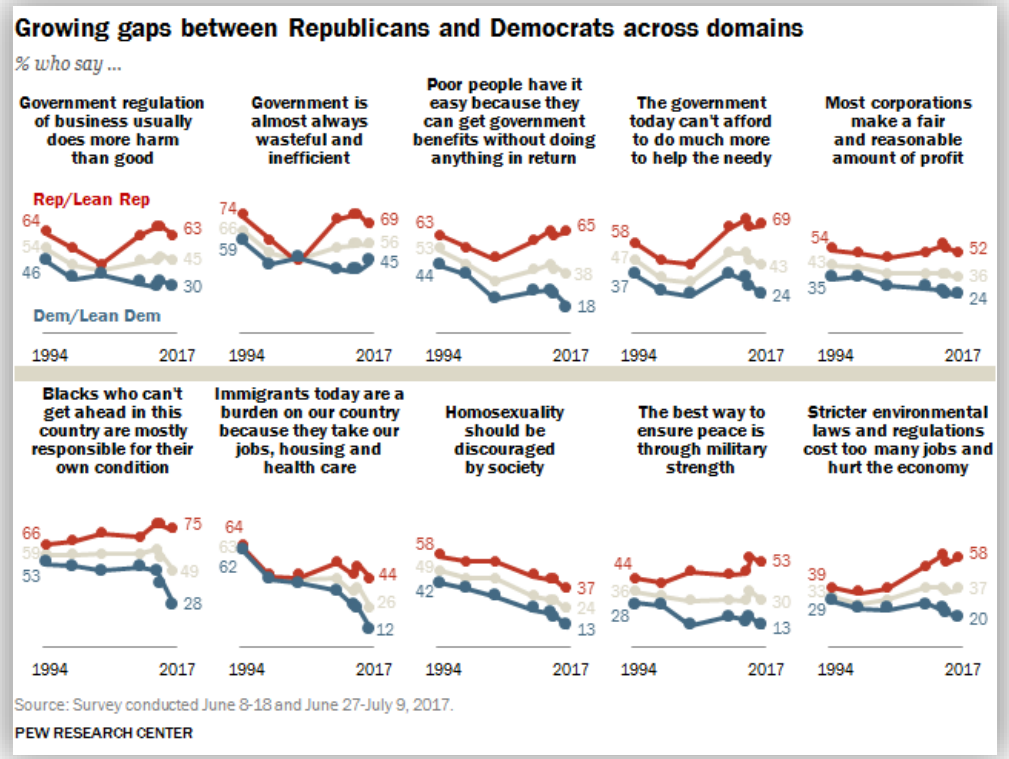
- Participation
- Increase in knowledge
- Expression



Lorenz-Spreen, P. *et al.* A systematic review of worldwide causal and correlational evidence on digital media and democracy. 2023

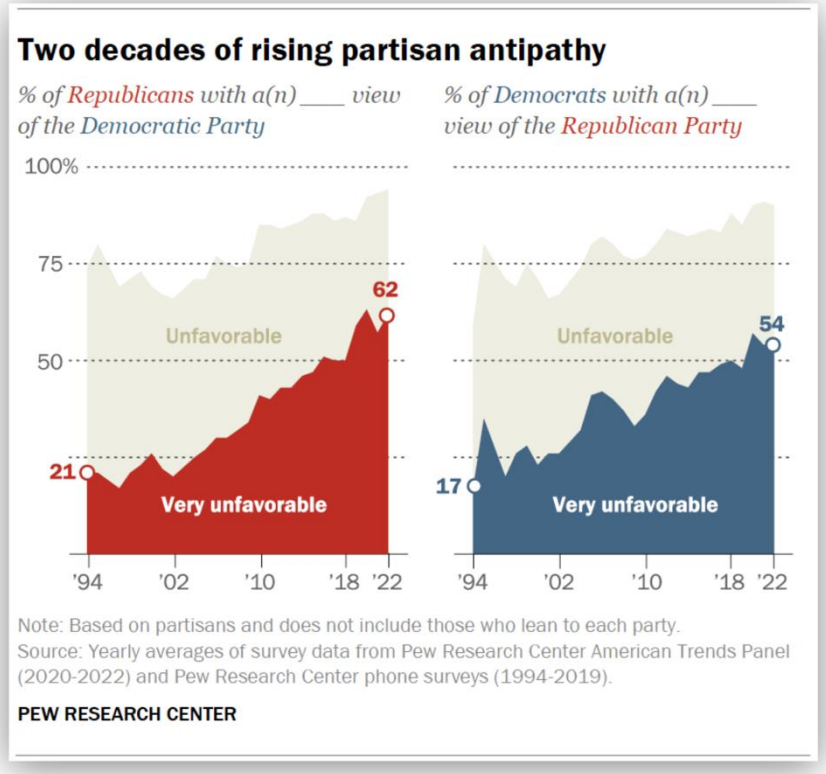
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Issue polarization

How far apart partisans are on policies



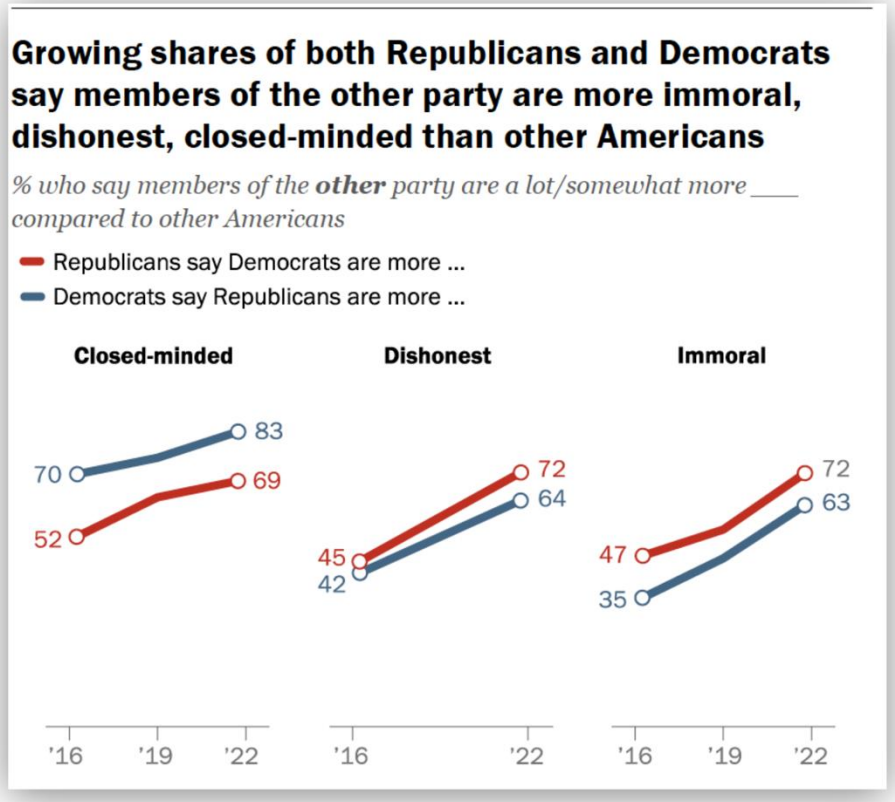
Affective polarization

The gap between warmth towards in- and out-group

Pew Research Center

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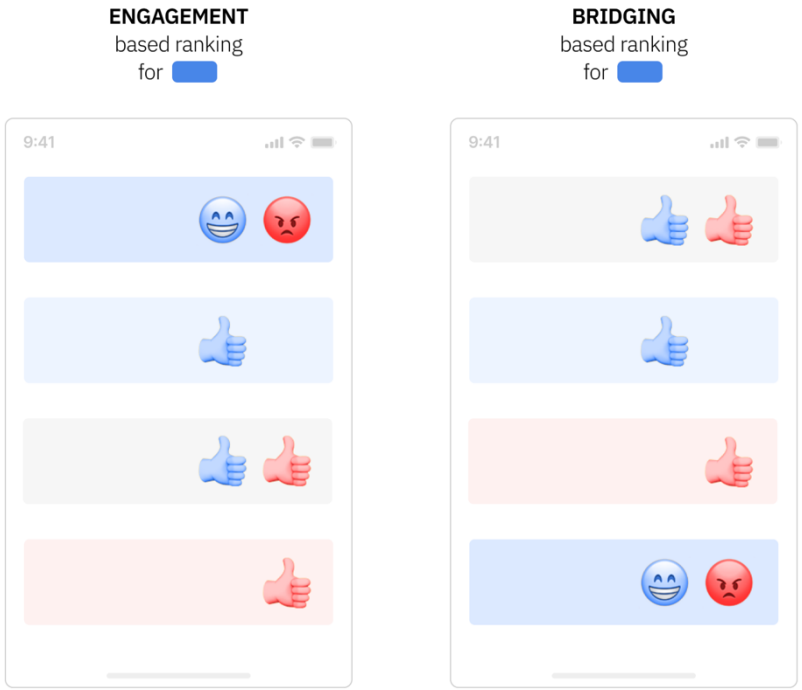


The consequence

- People overestimate ingroup homogeneity & outgroup extremity
- Anticipated conflict makes them more likely to avoid political discussion with outgroup members.

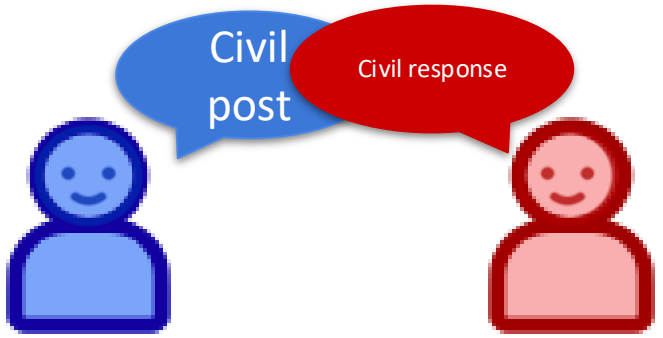
-> Feedback loop

How to promote prosocial content?

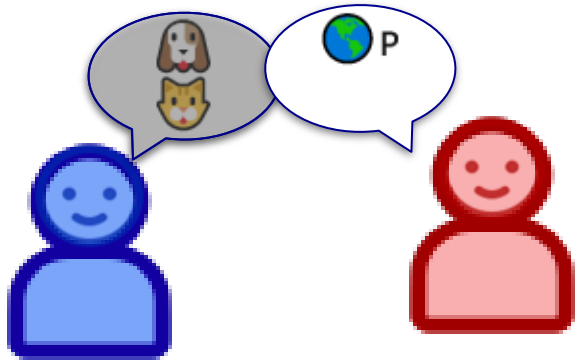


Diverse approval (signal based)

Toxicity resillience



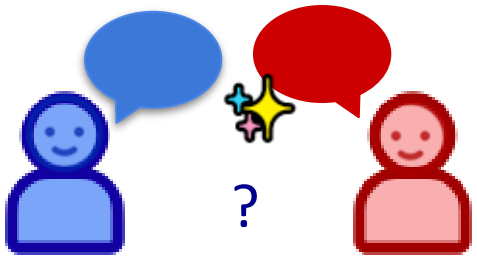
Controversy



Constructive conflict (content based)

Instead of relying on engagement, what can we do?

Design heuristic for constructive conflict



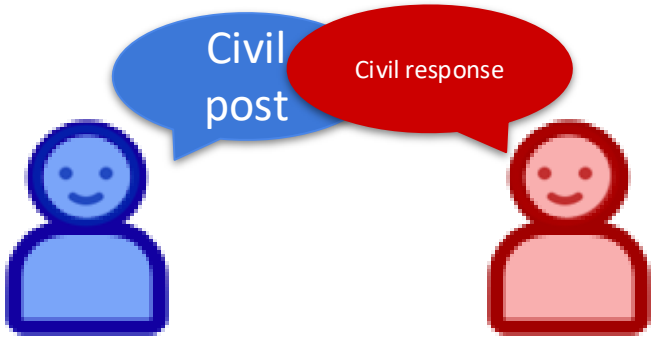
How to promote content that bridge people across divides?

- Can conversations about critical democratic issues be respectful?
- Are controversial political topics inherently prone to hostility?

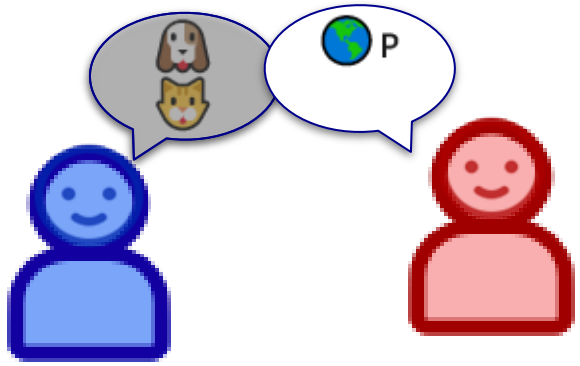
Constructive conflict

Definition: Respectful discussion on important topics that often spark debate among people

Toxicity resillience



Controversy



Seckin, O. C. et al. Identifying Constructive Conflict in Online Discussions through Controversial yet Toxicity Resilient Posts. 2026.

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Constructive conflict

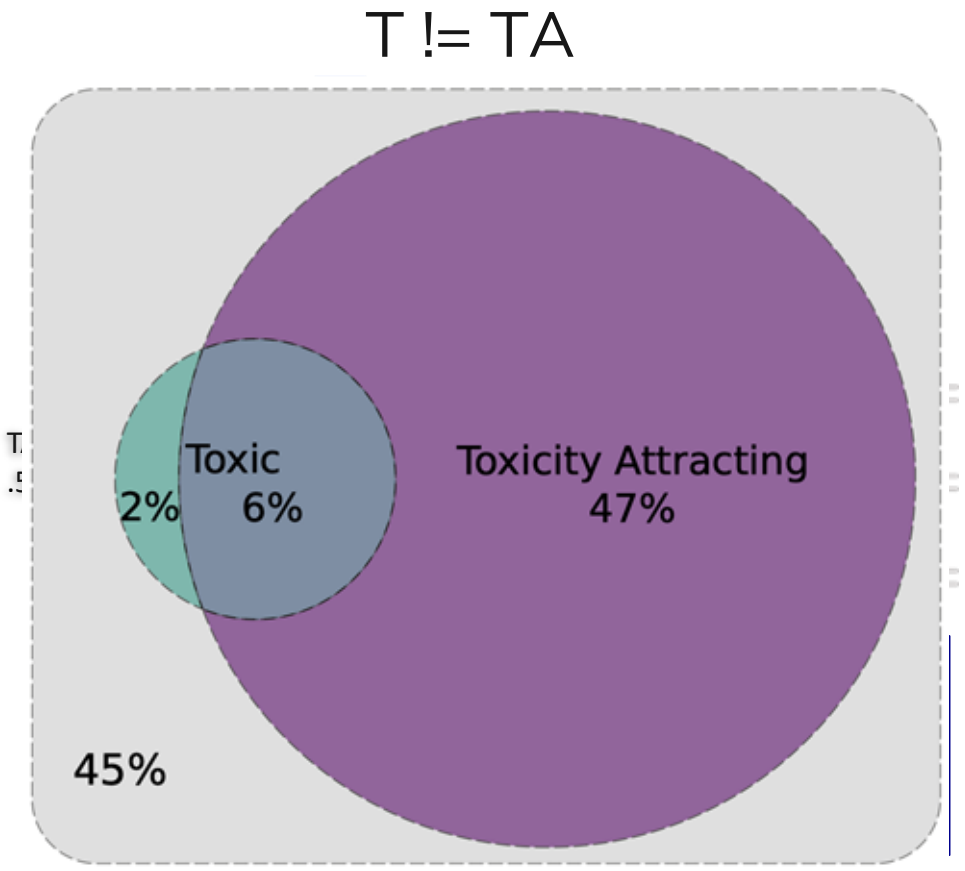
Respectful discussion on important topics that often spark debate among people

Toxicity resillience

Is this post likely to attract toxic responses?

Regression model to predict Toxicity Attraction scores (TA) — the average toxicity score of comments

- Reddit data (2022); 1.5M submissions; 12.9M comments across 50 subreddits (politics, gaming, dating…)
- Pretrained DistilBERT model with a regression output layer, MSE loss function
- 70-15-15% split for training, validation, testing
- Spearman correlation of 0.7 on the test set



Seckin, Truong, Flammini & Menczer, ICWSM R&R (2025)

Seckin, O. C. et al. Identifying Constructive Conflict in Online Discussions through Controversial yet Toxicity Resilient Posts. 2026.

Constructive conflict

Respectful discussion on important topics that often spark debate among people

C ∈ [0,1]
0: not controversial
1: highly controversial

C=0 “Snow leopard”, “copyright”, “theater”…
C=0.5 “Michael Jackson's health and appearance”, “Jehovah”, “1956 Winter Olympics”…
C=1 “Israel, Palestine, and the United Nations”, “Feminism”, “Pornography”

Controversy

Is this a topic people are likely to argue about?

Regression model to predict Controversiality score

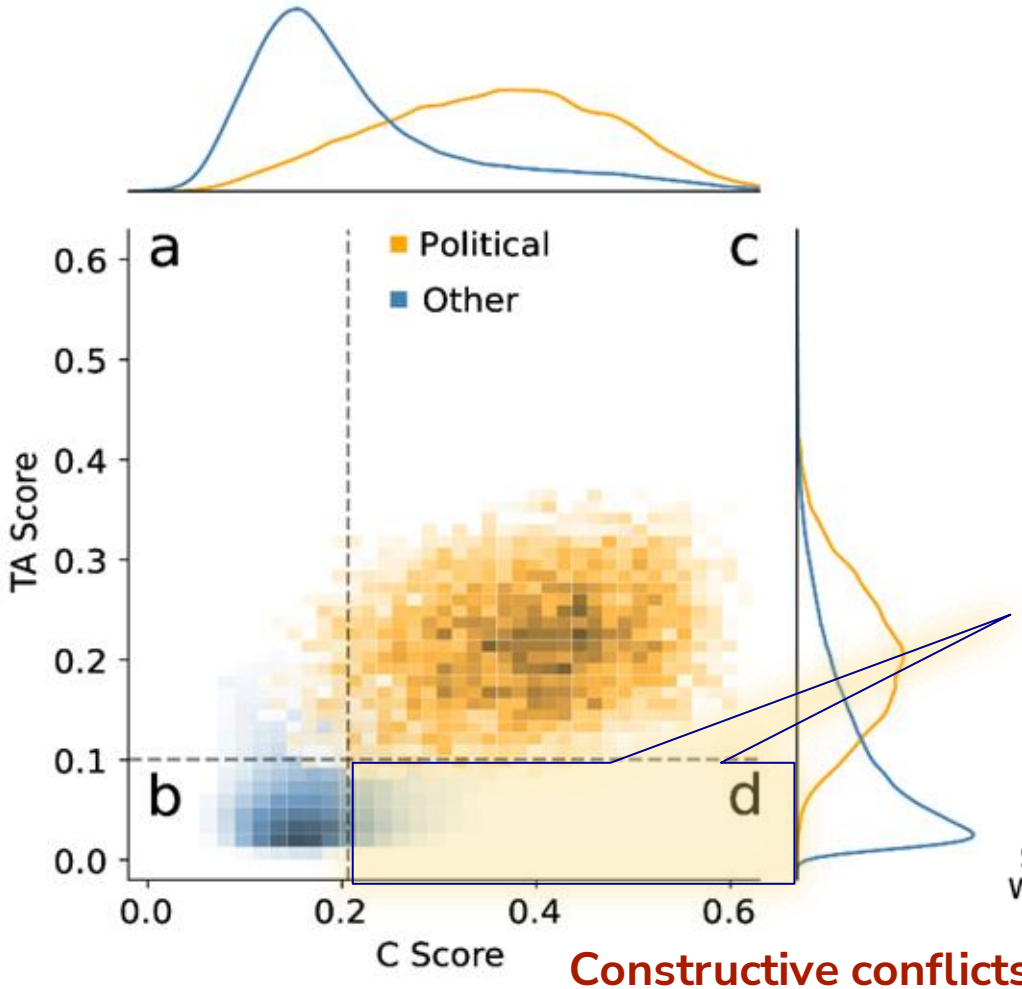
- Wikipedia dataset
  - 3,560 concepts and their description
  - Manually annotated by 10 humans
- Pretrained DistilBERT model with a regression output layer, MSE loss function
- 70-15-15% split for training, validation, testing
- Spearman correlation of 0.76 on the test set

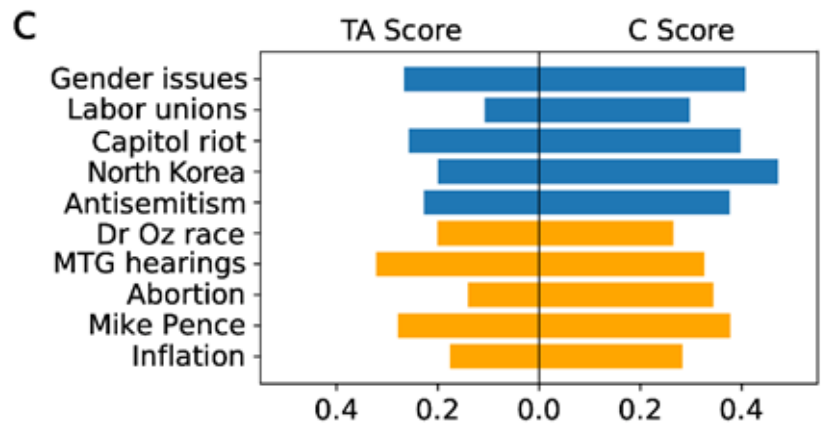
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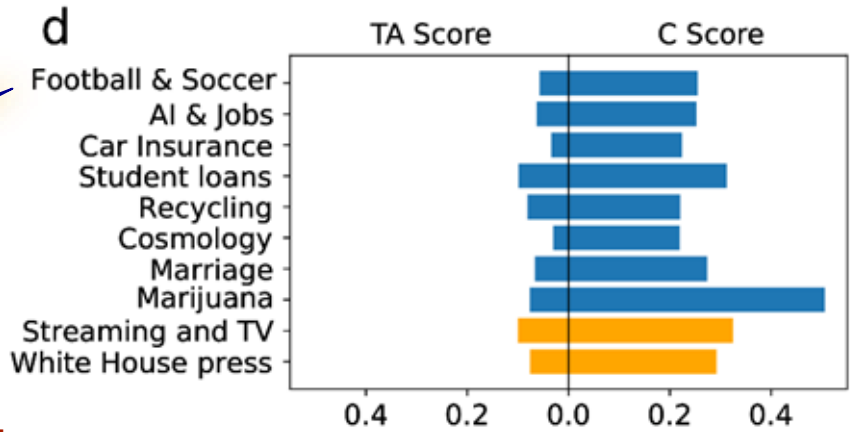
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The relationship between Civility and Controversy







Constructive conflicts exist!

Seckin, O. C. et al. Identifying Constructive Conflict in Online Discussions through Controversial yet Toxicity Resilient Posts. 2026.

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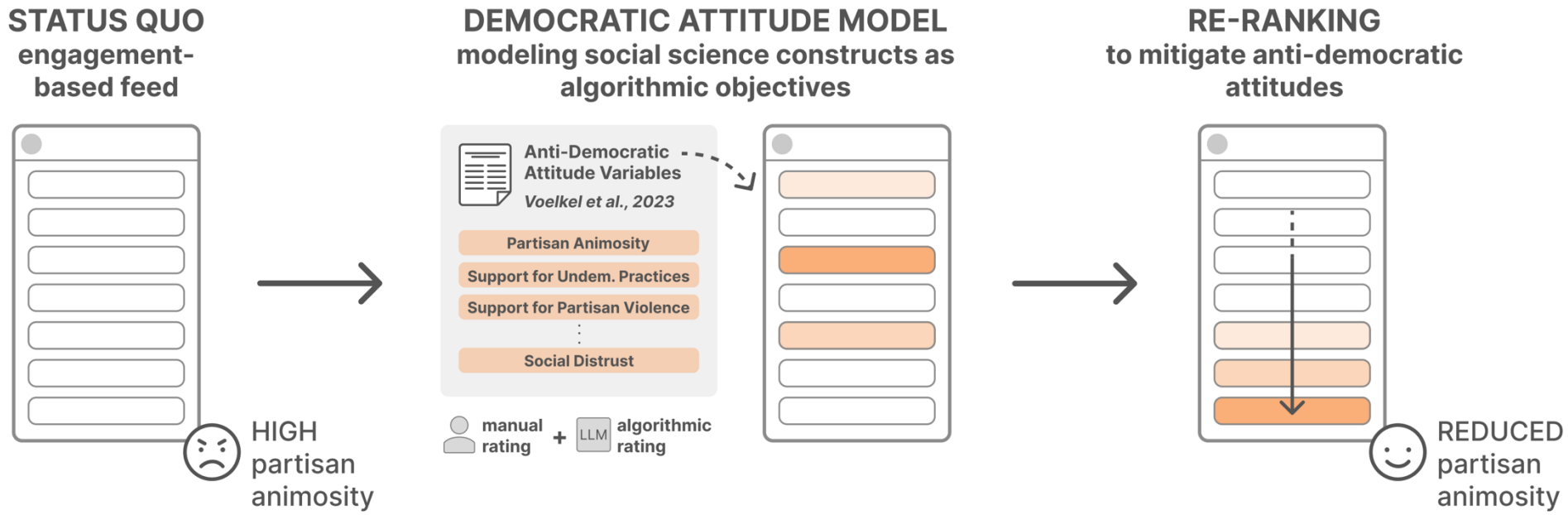
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Examples

Topic	Submission	TA %	C %
Abortion	VP Kamala Harris continues to tell Christians they should support the killing of pre-born humans.	99	91
	Should abortion be limited? I'm asking if abortion should be limited around 12-15 weeks? By then the fetus has a heartbeat, nervous system, functional organs, and yes eyes and mouth. Also women have bumps by then too. 99% of women know they are pregnant before 12 weeks and can make decision. I don't think abortion should be totally banned but 15 week is a reasonable compromise.	39	96
Healthcare	If the US was actually serious about supporting small businesses, they would enact universal healthcare. Working for a small business is effectively a death sentence, either financially or physically or both. Corporations hold the golden phallus of health care to dangle in front of your face. No health care only strengthens corporate asphyxiation on workers.	75	98
	Would making healthcare free in the U.S. be more expensive due to higher taxes, or cheaper due to no insurance payments/co-pays? I've been wondering this since the topic of "If we make healthcare free our taxes will go up". Although this is obviously true, I find that people forget they also wont be directly shilling out money every year for their insurance. So my question is in the title, I know taxes in other countries are a lot higher than in the U.S. because of that reason, but when incorporating the money you technically save from insurance and copayments would it still be that much of a difference?	10	88
Gun Laws	Enough is enough. Anyone who opposes gun control in this country is psychopathic unamerican scum.	95	98
	What are actual ways where gun laws can be re-established where both parties can be satisfied (whether for or against gun-control)?	30	99
LGBTQ+	Hating someone for being gay is a lifestyle choice.	99	84
	what's it called when you feel as if you were born the wrong gender but don't wanna go through with a transition of any sort because you know you wouldn't really be that gender? i in no way mean to invalidate anyone at all but i've been having some thoughts bout myself but i know it wouldn't be the same as being born as the opposite gender, i wanna look more into this so if someone could point me in the right direction of what to be looking up that would be great, and thank you in advance :)	33	51
Climate Change	This idiot still pushing climate change as a hoax. Same as he did with COVID.	92	72
	If global warming becomes a problem couldn't everyone just move to a cooler place Like say the temperatures get really hot Couldn't everyone just move to a place/country that is colder? Or for flooding couldn't everyone just move inland and be all good?	6	81

Seckin, O. C. et al. Identifying Constructive Conflict in Online Discussions through Controversial yet Toxicity Resilient Posts. 2026.

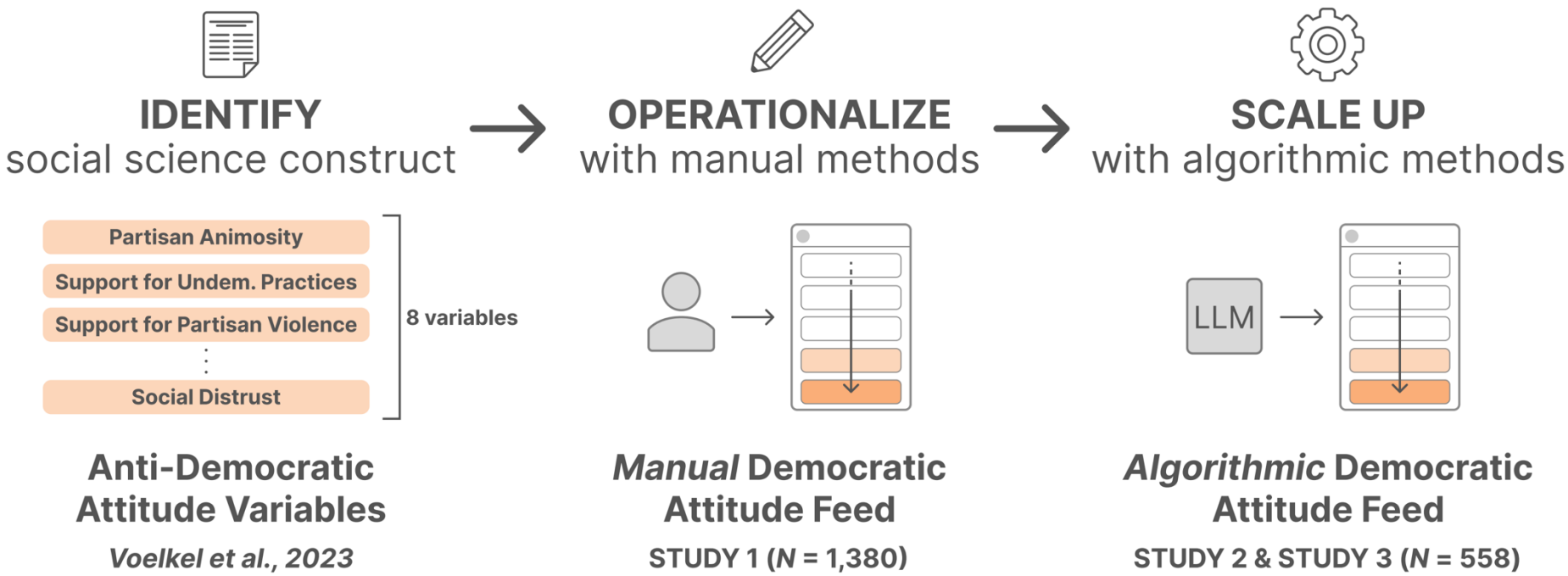
How do we make feeds that align with user values?



Embedding Democratic Values into Social Media AIs via Societal Objective Functions
Jia et al., 2024

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IDENTIFY
social science construct
OPERATIONALIZE
with manual methods
SCALE UP
with algorithmic methods
Partisan Animosity
Support for Undem. Practices
Support for Partisan Violence
Social Distrust
8 variables
LLM
Anti-Democratic Attitude Variables
Voelkel et al., 2023
Manual Democratic Attitude Feed
STUDY 1 (N = 1,380)
Algorithmic Democratic Attitude Feed
STUDY 2 & STUDY 3 (N = 558)

How do we make feeds that align with user values?

